a) Suppose we are given the following table definitions with certain records in each table. (Underline attribute represent primary key attributes.).

EMPLOYEE (EID, NAME, POST, AGE)

POST (POST-TITLE, SALARY)

PROJECT (PID, PNAME, DURATION, BUDGET)

WORK-IN (PID, EID, JOIN-DATE)

Write the SQL statement for

i. List the name of employees whose age is greater than the average age of all employees.

ii. Display all employee numbers of those employee who are not working in any project

iii. List name of employee and their salary who are working in the project "DBMS".

iv. Update the database so that "Rishab" now lives in "Butwal".

b) Consider the following relations:

Employee (emplD, FirstName, LastName, address, DOB, sex, position, deptNo)

Department (dtptNo, deptName, mgr, empID)

Project (projNo, projName, deptNo)

Work on (empID, projNo, hours worked)

Write the SQL statements for the following:

i. List the name and addresses of all employees who work for the IT department.

ii. List the total hours worked by each employee, arranged in order of department number and within department, alphabetically by employee surname.

iii. List the total number of employees in each department for those departments with more than 10 employees.

iv. List the project number, project name and the number of employees who work on that project.

c) Consider the following base relvars (relations):

employee (emp_name, street, city)

works (emp_name, company, salary)

company (comp_name, city)

manages (emp_name, mngr_name)

Give the SQL statements for each of the following:

i. Find employee names that lives in the city same as the company city

ii. List employee detail who earns more than Rs. 17,000

iii. Update address of Kiran to Sanepa, Lalitpur

iv. Create view for which employee earns Rs. 18,000 or more

v. Delete all employees for which manager name is null

d) Write SQL statements for the following queries in reference to relation **Emp_time** provided.

Eid#	Name	Start_time	End_time
E101	Mangale	10:30	18:30
E102	Malati	8:30	14:30
E103	Fulmaya	9:00	18:00

i. Create the table **Eid#** as primary key and insert the values provided.
ii. Display the name of the employee whose name start from letter 'M' and who work for more than seven hours.
iii. Delete the entire contents of the table so that new data can be inserted.

e) Consider the employee database of the following figure, where the primary keys are underlined. Give an expression in SQL for each of the following queries.

employee (<u>employee-name,</u> street, city)

works (<u>employee-name,</u> company-name, salary)

company (<u>company-name,</u> city)

manages (<u>employee-name,</u> manager-name)

1. Find the names and cities of residence of all employees who worked for NBL
2. Find the names, street address and cities of residence of all employees who worked for NBL and earn more than Rs.10,000
3. Find all the employees in the database who do not work for NBL
4. Find all employees in the database who earn more than each employee of NBB
5. Give all managers of First Bank Corporation a 10 percent raise unless the salary becomes greater than Rs.100,000; in such case.

f) Write SQL statements for the following queries in reference to relation **Emp_time** provided.

Eid*	**Name**	**Start_time**	**End_time**
E101	Hari	10:15	18:00
E102	Malati	8:00	15:30
E103	Kalyan	9:30	17:00

i. Create the table **Eid*** as primary key and insert the values provided.

ii. Display the name of the employee whose name start from letter 'M' and who work for more than seven hours.

iii. Delete the entire contents of the table so that new data can be inserted.

g) Consider the following relational database.

job (job-id, job-title, minimum-salary, maximum-salary)

employee (emp-no, emp-name, emp-salary, dept-no)

deposit (acc-no, cust-name, branch-name, amount, account-date)

borrow (loan-no, cust-name, branch-name, amount)

department (dept-no, dept-name)

Write the SQL statements for the following queries.

1. Give name of depositors whose branch name starts from 'D'.
2. Give employee name(s) whose salary is between Rs. 20000 to 30000 and department name is Finance.
3. Delete borrower details whose amount is less than 25000.
4. Insert new employee details {'E105', 'Roshan Singh', '60000', 'D109'}.

h) Consider the following relational database.

Project (P_ID, P_Name, P_Location, Type)

Employee (Emp_ID, Emp_Name, Address, Salary, Post, Date_join)

Works (Dept_No, Emp_ID, Shift)

Write the SQL statements for the following queries.

i. Insert new project {'P105', 'The Helping Hands', 'Nagbazar', 'Private'}.

ii. List the name of the employees whose salary is above the average salary.

iii. Remove record of all employees who work in morning shift at project located in New Baneshewor, Kathmandu.

iv. Change address and post of employee “Ramesh” to “Pokhara” and “Lecturer”.

i) Consider the following relational database.

EMPLOYEE(<u>EMPNO</u>, NAME, ADDRESS)

PROJECT(<u>PNO</u>,PNAME)

WORKON(EMPNO, PNO)

PART(<u>PARTNO</u>, PARTNAME, QTY_ON_HAND)

USE(<u>EMPNO,PNO,PARTNO</u>,NUMBER)

Write the SQL statements for the following queries.

5. To create the EMPLOYEE table. (assume your own data types for attributes)
6. Give details of employee whose name ends with ‘H’.
7. To retrieve the name of the parts for which the number of parts used is greater than 80.
8. To insert a record into the EMPLOYEE table.